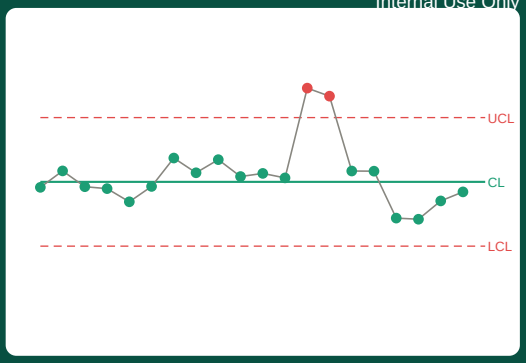
QUALITY & PROCESS

Statistical Process Control

Monitor process outputs using control charts to detect changes in process behaviour early — before they produce



WHEN TO USE

In any repetitive process where early detection of change enables prevention rather than detection and rework.

HOW TO APPLY — STEPS

1. Identify the critical-to-quality characteristic
2. Select the appropriate chart type for the data
3. Establish control limits from a period of stable process operation
4. Plot results in real time as the process runs
5. React to signals (points outside limits, runs, trends) with structured investigation

PRACTITIONER TIP

Control limits come from process data. Specification limits come from customer requirements. They are NEVER the same thing.

KEY OUTPUTS

- Real-time control charts
- Process capability summary
- Reaction plan for each signal type

HOW TO MEASURE SUCCESS

False alarm rate; time to detect a genuine process shift; Cpk improvement trend

TOOL CATEGORY

Quality & Process